

MIAE 221: Materials Science for Engineers

Part 8: Phase Diagrams & Iron-Carbon Systems Master Guide

Executive Overview & Core Engineering Principles

A pure element has a fixed melting temperature and uniform properties. However, virtually all structural engineering alloys—from the titanium skins of high-performance aircraft to the reinforced structural steels of high-rise skyscrapers—are multiphase mixtures composed of two or more chemical components.

A **Phase Diagram** (also called an **equilibrium diagram** or **constitution diagram**) is a graphical map showing the relationships between environmental constraints (temperature, pressure) and composition, indicating the exact phases present in thermodynamic equilibrium.

Mastering phase diagrams enables an engineer to:

1. Predict what microstructural phases form when an alloy solidifies or cools slowly.
 2. Determine the exact chemical compositions of each coexisting phase using a horizontal **tie line**.
 3. Calculate the precise mass/weight fractions of each phase using the **inverse lever rule**.
 4. Understand and control invariant reactions (**eutectic**, **eutectoid**, **peritectic**).
 5. Master the **Iron-Carbon (Fe-Fe₃C) system**, the technological backbone of modern civilization.
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1. Thermodynamic Definitions & The Gibbs Phase Rule

Foundational Terminology

- **Component (*C*)**: A chemically independent constituent (pure element or stoichiometric chemical compound) that composes the system (e.g., in water-salt: H₂O and NaCl; in brass: Cu and Zn; in carbon steel: Fe and C).
- **System**: A specific body of material under consideration.
- **Phase (*P*)**: A physically distinct, chemically homogeneous, and mechanically separable portion of a system with uniform crystal structure and atomic bonding throughout (e.g. solid α , liquid *L*, solid Fe₃C).
- **Solubility Limit**: The maximum concentration of solute atoms that can dissolve into a solvent matrix at a given temperature to form a single-phase solid solution. Adding solute beyond this limit causes a second distinct phase to precipitate.

The Gibbs Phase Rule

Formulated by J. Willard Gibbs, the phase rule governs the thermodynamic relationship between the number of coexisting phases (*P*), the number of components (*C*), and the number of independent externally controllable

variables (degrees of freedom F):

$$P + F = C + N$$

Where N is the number of external non-compositional variables (normally Temperature and Pressure, so $N = 2$).

The Condensed Gibbs Phase Rule for Metallurgy

In practical metallurgical and materials engineering, phase transformations occur at a constant atmospheric pressure of 1 atm. Because pressure is held fixed, one degree of freedom is consumed, reducing the equation to the **condensed phase rule**:

$$P + F = C + 1$$

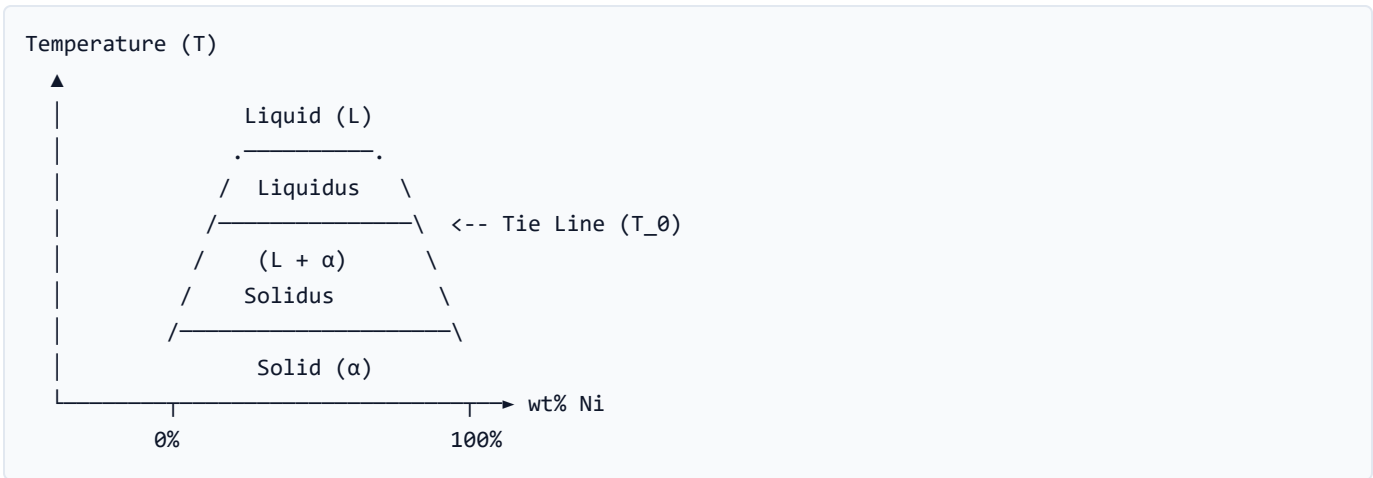
For a **binary alloy system** ($C = 2$):

$$P + F = 2 + 1 = 3 \implies F = 3 - P$$

Number of Phases (P)	Degrees of Freedom (F)	Thermodynamic Classification	Physical Meaning
$P = 1$ (Single Phase)	$F = 2$	Bivariant	Both Temperature and Composition can be independently varied within the single-phase field without altering the phase state.
$P = 2$ (Two Phases)	$F = 1$	Univariant	Choosing Temperature automatically fixes the equilibrium compositions of both coexisting phases at the tie-line endpoints!
$P = 3$ (Three Phases)	$F = 0$	Invariant	Three phases coexist at only one unique, invariant temperature and unique invariant compositions (e.g., the eutectic point).

2. Binary Isomorphous Systems & The Lever Rule

An **isomorphous system** exhibits complete liquid and solid solubility across the entire composition range (0 to 100 %). The classic example is the **Copper-Nickel (Cu-Ni)** system, because both Cu and Ni satisfy all four Hume-Rothery rules (both FCC, atomic radius difference < 2, electronegativities 1.9 vs 1.8, same valency).



Key Phase Boundaries

- **Liquidus Line:** The boundary line above which only liquid exists. Solidification begins when cooling through this line.
- **Solidus Line:** The boundary line below which the alloy is completely solid. Melting begins when heating through this line.
- **Two-Phase Region ($L + \alpha$):** Between the liquidus and solidus lines, liquid and solid α coexist in equilibrium.

The Tie-Line & Inverse Lever Rule Algorithm

When an alloy of overall composition C_0 is held at a temperature T_0 located within a two-phase region (e.g. $L + \alpha$), two fundamental questions must be answered:

1. **What are the compositions of the coexisting phases?**
2. **What are the relative mass fractions of each phase?**

Step 1: Draw the Horizontal Tie-Line (Isotherm)

Draw a horizontal line across the two-phase region at temperature T_0 , terminating at the adjacent single-phase boundary lines.

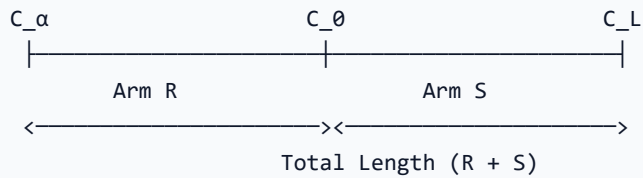
Step 2: Determine Phase Compositions (C_L and C_α)

Drop perpendicular vertical lines from the tie-line intersections to the horizontal composition axis:

- The intersection with the liquidus gives the liquid composition: C_L .
- The intersection with the solidus gives the solid composition: C_α .

Step 3: Compute Phase Weight Fractions via the Inverse Lever Rule

The overall tie-line acts as a mechanical lever balanced at the alloy's overall composition C_0 :



By conservation of mass:

- The mass fraction of the phase on the left (α) is proportional to the length of the **opposite** tie-line arm on the right ($S = C_L - C_0$):

$$W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{S}{R + S}$$

- The mass fraction of the phase on the right (L) is proportional to the length of the **opposite** tie-line arm on the left ($R = C_0 - C_\alpha$):

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{R}{R + S}$$

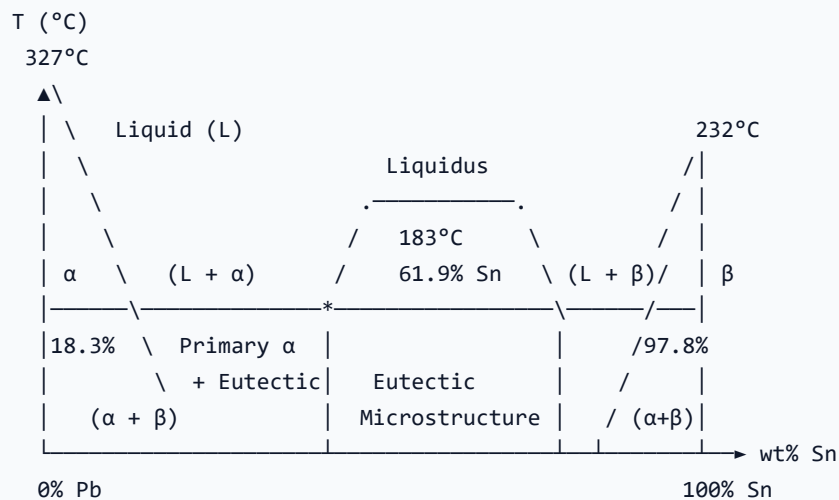
- As a mandatory mathematical check:

$$W_\alpha + W_L = 1.00 \quad (100$$

3. Binary Eutectic Systems (Pb-Sn Phase Diagram)

In systems where components have limited solid solubility in one another (due to differences in atomic size, valence, or crystal structure), an intermediate invariant reaction occurs: the **eutectic reaction**.

The classic engineering benchmark is the **Lead-Tin (Pb-Sn)** soldering system:



Foundational Invariant Points in Pb-Sn

- **Eutectic Temperature:** $T_E = 183^\circ\text{C}$.
- **Eutectic Composition:** $C_E = 61.9 \text{ wt}$.
- **Maximum Solid Solubility of Sn in Lead (α):** $C_{\alpha,\text{max}} = 18.3 \text{ wt}$ at 183°C .
- **Maximum Solid Solubility of Pb in Tin (β):** $C_{\beta,\text{max}} = 97.8 \text{ wt}$ (2.2 wt) at 183°C .

The Invariant Eutectic Reaction

Upon cooling through 183°C , liquid of composition 61.9 wt solidifies isothermally into two distinct solid phases:

$$L(61.9 \text{ wt})$$

Why Does the Eutectic Microstructure Form Alternating Lamellae?

Because solid α is rich in lead (81.7 wt) while solid β is rich in tin (97.8 wt), solidification requires rapid redistribution of atoms. Atoms diffuse sideways over minimal distances ahead of the advancing solidification interface, producing a characteristic **alternating lamellar (layered) structure** of α and β plates.

Primary (Proeutectic) Phase vs. Eutectic Microconstituent

For an alloy that is **hypoeutectic** ($C_0 < C_E$, e.g. 40 wt):

1. **Between Liquidus and 183°C :** Primary (proeutectic) α dendrites nucleate and grow from the liquid. The remaining liquid becomes increasingly enriched in Sn along the liquidus line until it reaches 61.9 wt.
2. **At 183°C :** All remaining liquid freezes into the lamellar eutectic structure ($\alpha + \beta$).
3. **Below 183°C :** The resulting room-temperature microstructure contains two distinct **microconstituents**:
 - **Primary (Proeutectic) α' :** Large rounded dendritic grains.
 - **Eutectic Mixture:** Lamellar colonies of fine alternating α and β plates.

Calculating Microconstituent Fractions (Lever Rule Just Above 183°C)

To find the amount of primary α' microconstituent (rather than total α phase), apply the lever rule at $T = 183^\circ\text{C} + \Delta T$, where the phases in equilibrium are primary α' and liquid of eutectic composition $L(C_E)$:

$$W_{\alpha'} = \frac{C_E - C_0}{C_E - C_\alpha} = \frac{61.9 - C_0}{61.9 - 18.3}$$

The mass fraction of the eutectic microconstituent is simply the fraction of liquid that transformed:

$$W_{\text{eutectic}} = 1 - W_{\alpha'} = \frac{C_0 - C_\alpha}{C_E - C_\alpha} = \frac{C_0 - 18.3}{61.9 - 18.3}$$

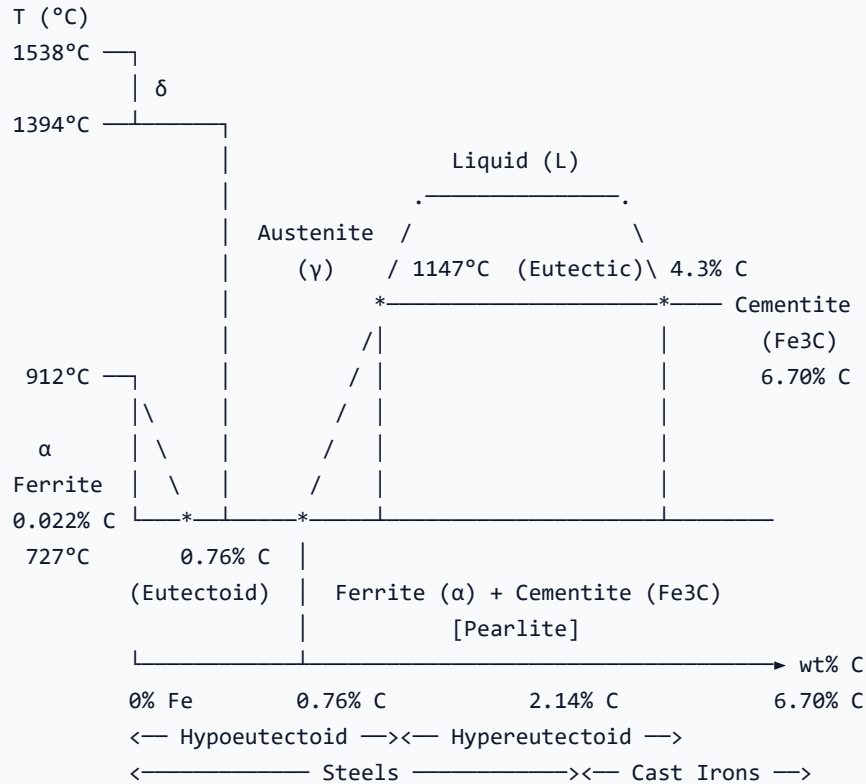
4. Classification of Invariant Three-Phase Reactions

An **invariant reaction** occurs at a unique temperature and composition ($F = 0$) where three phases coexist in equilibrium. The classical reactions are classified upon cooling:

Reaction Name	Reaction Equation (Upon Cooling)	Key Feature	Classic Example
Eutectic	$\text{Liquid} \xrightarrow{\text{cool}} \text{Solid}_1 + \text{Solid}_2$	Single liquid freezes into two distinct solids.	Pb-Sn at 183°C (61.9) Fe-C at 1147°C (4.3)
Eutectoid	$\text{Solid}_1 \xrightarrow{\text{cool}} \text{Solid}_2 + \text{Solid}_3$	Single solid decomposes into two distinct solids.	Fe-Fe ₃ C at 727°C (0.76)
Peritectic	$\text{Liquid} + \text{Solid}_1 \xrightarrow{\text{cool}} \text{Solid}_2$	Liquid reacts with a solid to form a new single solid.	Fe-C at 1493°C (0.16)
Peritectoid	$\text{Solid}_1 + \text{Solid}_2 \xrightarrow{\text{cool}} \text{Solid}_3$	Two distinct solids react to form a new single solid.	Cu-Sn bronze system
Monotectic	$\text{Liquid}_1 \xrightarrow{\text{cool}} \text{Liquid}_2 + \text{Solid}_1$	A liquid decomposes into a different liquid and a solid.	Cu-Pb bearing alloys

5. The Iron-Carbon (Fe-Fe₃C) Phase Diagram

The iron-carbon system forms the metallurgical bedrock of structural engineering. Steels contain between 0.02 wt and 2.14 wt, while cast irons contain between 2.14 wt and 6.70 wt.



The Three Allotropes of Iron

1. α -Ferrite (BCC):

- Stable from room temperature up to 912°C.
- **Extremely low carbon solubility:** Maximum 0.022 wt at 727°C, dropping to < 0.005 wt at room temperature!
- Mechanical properties: Soft, ductile, ferromagnetic.

2. γ -Austenite (FCC):

- Stable from 912°C to 1394°C.
- **Substantially higher carbon solubility:** Maximum 2.14 wt at 1147°C!
- Mechanical properties: Non-magnetic, highly formable.

3. δ -Ferrite (BCC):

- Stable at high temperatures (1394°C to melting point 1538°C).

[!IMPORTANT] Why Does Austenite (FCC) Dissolve 100× More Carbon than Ferrite (BCC)?

- Although BCC has a lower atomic packing factor (0.68 vs 0.74), its interstitial voids are fragmented into small, severely distorted sites ($r_{\text{site}} \approx 0.036 \text{ nm}$). Inserting a carbon atom ($r_C \approx 0.071 \text{ nm}$) causes severe tetragonal lattice strain.
- In FCC austenite, the octahedral interstitial voids located at cube edge centers and body center are symmetrical and significantly larger ($r_{\text{site}} \approx 0.053 \text{ nm}$), accommodating carbon with far lower strain energy!

Cementite (Fe_3C)

- A metastable intermediate intermetallic compound containing exactly 6.70 wt (25 at).
- Crystal structure: Orthorhombic.
- Properties: **Extremely hard and brittle**, providing the primary strengthening microconstituent in steels.

The Invariant Eutectoid Reaction (727°C, 0.76 wt)

Upon slow cooling through 727°C, austenite of eutectoid composition transforms into **pearlite**:



- **Pearlite** is NOT a single phase! It is a two-phase microconstituent consisting of alternating parallel lamellae of ductile α -ferrite (approx. 88 wt) and hard Fe_3C cementite (approx. 12 wt).

Hypoeutectoid vs. Hypereutectoid Steels

- **Hypoeutectoid Steels** ($C_0 < 0.76 \text{ wt}$):
 - Slow cooling below the A_3 line produces **proeutectoid ferrite** (α') along austenite grain boundaries.
 - Below 727°C, the remaining austenite transforms into pearlite.
 - Final microstructure: **Proeutectoid ferrite + Pearlite**.
- **Hypereutectoid Steels** ($0.76 < C_0 < 2.14 \text{ wt}$):

- Slow cooling below the A_{cm} line produces a network of hard, brittle **proeutectoid cementite** ($\text{Fe}_3\text{C}'$) along grain boundaries.
- Below 727°C , remaining austenite transforms into pearlite.
- Final microstructure: **Proeutectoid cementite + Pearlite**.

6. Fully Solved Master Quantitative Archetypes

Problem 1: Pb-Sn Lever Rule at 150°C in Two-Phase Solid Region (Slide 30)

Problem Statement: An alloy of 40 wt ($C_0 = 40$) is cooled to 150°C . At 150°C , the tie line intersects the solid solubility curves at:

- $C_\alpha = 11 \text{ wt}$
- $C_\beta = 99 \text{ wt}$

Determine:

1. The phases present in equilibrium.
2. The chemical compositions of each phase.
3. The relative weight fractions (W_α and W_β).

Step-by-Step Solution:

1. Phases Present: The state point ($T = 150^\circ\text{C}$, $C_0 = 40$) lies in the two-phase field: $\alpha + \beta$.

2. Compositions of Phases: Read directly from the tie-line endpoints:

- $C_\alpha = 11 \text{ wt}$ (89 wt)
- $C_\beta = 99 \text{ wt}$ (1 wt)

3. Weight Fractions (Inverse Lever Rule): Total tie-line length:

$$L = C_\beta - C_\alpha = 99 - 11 = 88$$

Fraction of α (opposite right arm): $W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{99 - 40}{99 - 11} = \frac{59}{88} \approx \mathbf{0.6705} \quad (\mathbf{67.0\text{ wt}\%})$

Fraction of β (opposite left arm): $W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{40 - 11}{99 - 11} = \frac{29}{88} \approx \mathbf{0.3295} \quad (\mathbf{33.0\text{ wt}\%})$

Check: $67.0. \checkmark$

Problem 2: Pb-Sn Lever Rule at 220°C in Solid + Liquid Region (Slide 31)

Problem Statement: A 40 wt alloy is held at 220°C . At this temperature, the horizontal tie line intersects the solidus at $C_\alpha = 17 \text{ wt}$ and the liquidus at $C_L = 46 \text{ wt}$. Calculate the weight fractions of solid α and liquid L .

Step-by-Step Solution:

Total tie-line length:

$$L = C_L - C_\alpha = 46 - 17 = 29$$

Weight fraction of liquid W_L : $W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{40 - 17}{46 - 17} = \frac{23}{29}$
 $\approx 0.7931 \quad (\mathbf{79.3\%})$

Weight fraction of solid α W_α : $W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{46 - 40}{46 - 17} = \frac{6}{29}$
 $\approx 0.2069 \quad (\mathbf{20.7\%})$

Problem 3: Hypoeutectoid Steel Microconstituent Calculation (Slide 68)

Problem Statement: A hypoeutectoid plain carbon steel containing 0.40 wt is slowly cooled to just below the eutectoid temperature (727°C). Given: $C_\alpha = 0.022$ wt, $C_{\text{eutectoid}} = 0.76$ wt, and $C_{\text{cementite}} = 6.70$ wt.

1. Calculate the weight fractions of **proeutectoid ferrite (α')** and **pearlite (p)**.
2. Calculate the **total weight fraction of ferrite (α)** in the steel.

Step-by-Step Solution:

Part 1: Proeutectoid Ferrite vs. Pearlite Fractions Apply the lever rule just above 727°C between α (0.022) and austenite of eutectoid composition γ (0.76): $W_{\alpha'} = \frac{C_{\text{eutectoid}} - C_0}{C_{\text{eutectoid}} - C_\alpha} = \frac{0.76 - 0.40}{0.76 - 0.022} = \frac{0.360}{0.738} \approx 0.4878 \quad (\mathbf{48.8\%})$

The pearlite fraction equals the fraction of austenite that transforms: $W_p = W_\gamma = 1 - W_{\alpha'} = \frac{0.40 - 0.022}{0.738} = \frac{0.378}{0.738} \approx 0.5122 \quad (\mathbf{51.2\%})$

Part 2: Total Ferrite Phase Fraction Apply the lever rule across the entire phase diagram between α (0.022) and cementite (6.70): $W_{\alpha, \text{total}} = \frac{C_{\text{cementite}} - C_0}{C_{\text{cementite}} - C_\alpha} = \frac{6.70 - 0.40}{6.70 - 0.022} = \frac{6.30}{6.678} \approx 0.9434 \quad (\mathbf{94.3\%})$

Notice the crucial difference: Total ferrite is **94.3**, but proeutectoid ferrite is only **48.8**. The remaining **45.5** of ferrite is packaged inside the pearlite lamellae!

7. Exam Pitfalls & High-Yield Summary Table

Concept	Governing Formula	Key Mechanism	Common Exam Pitfall
Phase Composition	Endpoints of tie line	Horizontal line across two-phase region.	Confusing phase composition (C_L) with phase fraction (W_L).
Inverse Lever Rule	$W_A = \frac{\text{Opposite Arm}}{\text{Total Length}}$	Proportional to opposite segment.	Using adjacent segment instead of opposite arm!
Condensed Phase Rule	$P + F = C + 1$	Fixed 1 atm pressure consumes one variable.	Using $C + 2$ (standard gas rule) instead of $C + 1$.
Eutectic Reaction	$L \rightarrow \alpha + \beta$	Liquid freezes into two distinct solid phases.	Confusing eutectic ($L \rightarrow 2$ solids) with eutectoid.
Eutectoid Reaction	$\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$	Solid decomposes into two distinct solid phases.	Calling pearlite a single phase (it is a two-phase mixture).
Carbon Solubility	Austenite (2.14) vs Ferrite (0.022)	FCC octahedral voids are larger and symmetric.	Believing lower APF of BCC means larger interstitial holes.
Proeutectoid vs Total	Lever rule at 0.76 vs 6.70	Proeutectoid forms above 727°C ; pearlite forms at 727°C .	Conflating proeutectoid ferrite (α') with total ferrite (α).